SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I ATHIN SIVA.S_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CASE STUDY 1 – Smart Medical Diagnosis System

Given knowledge base:

- **R1: Fever(x) ∧ Cough(x) → Flu(x)**
- **R2: Fever(x) ∧ Rash(x) → Measles(x)**
- **R3: Cough(x) ∧ Breathlessness(x) → Pneumonia(x)**
- **Facts: Fever(PatientA) = True, Cough(PatientA) = True, Breathlessness(PatientA) = True.**

(a) Propositional representation and symbols

Let F = Fever(PatientA), C = Cough(PatientA), R = Rash(PatientA), B = Breathlessness(PatientA), L = Flu(PatientA), M = Measles(PatientA), and P = Pneumonia(PatientA). The rules can be written as:

- **F ∧ C → L**
- **F ∧ R → M**
- **C ∧ B → P**
- **Facts: F, C, B are true; R is not given.**

Here, ∧ means AND, → means implication, and ¬ means NOT. In the original first-order form, the rules are universally quantified, for example ∀x [Fever(x) ∧ Cough(x) → Flu(x)].

(b) Forward Chaining

Forward chaining starts with known facts and repeatedly applies rules whose premises are satisfied. It moves from facts toward conclusions.

- **Initial facts: Fever(PatientA), Cough(PatientA), Breathlessness(PatientA).**
- **Apply R1: Fever(PatientA) ∧ Cough(PatientA) are true, so derive Flu(PatientA).**
- **Apply R2: Fever(PatientA) is true, but Rash(PatientA) is not available. Therefore, Measles(PatientA) cannot be derived.**
- **Apply R3: Cough(PatientA) ∧ Breathlessness(PatientA) are true, so derive Pneumonia(PatientA).**
- **Final derived diagnoses: Flu(PatientA) and Pneumonia(PatientA).**

Therefore, the system identifies both Flu and Pneumonia as possible diagnoses from the given facts. Measles is not concluded because the required Rash fact is missing.

(c) Backward Chaining for Pneumonia(PatientA)

Goal: Pneumonia(PatientA). Backward chaining starts with the goal and searches for rules that can produce it.

- **Goal: Pneumonia(PatientA).**
- **Find a rule with conclusion Pneumonia(x): R3, Cough(x) ∧ Breathlessness(x) → Pneumonia(x).**
- **Substitute x = PatientA.**
- **New subgoals: Cough(PatientA) AND Breathlessness(PatientA).**
- **Both subgoals are directly present as facts in the KB.**
- **Therefore, both subgoals are true and the original goal Pneumonia(PatientA) is proved.**

Goal-reduction tree: Pneumonia(PatientA) ← Cough(PatientA) ∧ Breathlessness(PatientA) ← True ∧ True. Hence Pneumonia(PatientA) is verified.

CASE STUDY 2 – Autonomous Traffic Management Agent

Knowledge base:

- **R1: ∀x [Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)]**
- **R2: ∀x [ClearPath(x) → GreenSignal(x) ∧ RedSignal(x)]**
- **R3: ∀x∀y [Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)]**
- **Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance).**

(a) Unification with Rule 1

Rule 1 is Vehicle(x) ∧ EmergencyType(x) → ClearPath(x). The facts Vehicle(Ambulance) and EmergencyType(Ambulance) match the two premises. Therefore, the most general unifier is:

θ = {x/Ambulance}

Applying this substitution gives Vehicle(Ambulance) ∧ EmergencyType(Ambulance) → ClearPath(Ambulance). Since both premises are facts, ClearPath(Ambulance) is derived.

The same rule cannot currently be applied to CarA because EmergencyType(CarA) is not given.

(b) Resolution proof of Proceed(CarA)

First convert the relevant implications into clauses.

- **C1: ¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)**
- **C2a: ¬ClearPath(x) ∨ GreenSignal(x)**
- **C2b: ¬ClearPath(x) ∨ RedSignal(x)**
- **C3: ¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)**
- **Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA,Ambulance).**

To prove Proceed(CarA), use refutation by adding the negated goal ¬Proceed(CarA).

- **From C1 with Vehicle(Ambulance) and EmergencyType(Ambulance), derive ClearPath(Ambulance).**
- **From C2a and ClearPath(Ambulance), derive GreenSignal(Ambulance).**
- **Instantiate C3 with x = CarA and y = Ambulance: ¬Vehicle(CarA) ∨ ¬Behind(CarA,Ambulance) ∨ ¬GreenSignal(Ambulance) ∨ Proceed(CarA).**
- **Resolve with Vehicle(CarA), Behind(CarA,Ambulance), and GreenSignal(Ambulance). This gives Proceed(CarA).**
- **Resolve Proceed(CarA) with the negated goal ¬Proceed(CarA). The result is the empty clause □.**

Since the empty clause is obtained, the negated goal is inconsistent with the KB. Therefore, Proceed(CarA) is logically entailed.

(c) Can the KB handle two simultaneous emergency vehicles?

The current rules can derive ClearPath for every object satisfying both Vehicle(x) and EmergencyType(x). Therefore, if two emergency vehicles are explicitly represented by facts, the rules can independently derive ClearPath for both. Rule 2 can then derive signals for both.

However, the KB is not sufficient for realistic simultaneous emergency handling because it has no rule for prioritization, collision avoidance, lane allocation, timing, or conflicting traffic commands. Also, Rule 2 derives both GreenSignal(x) and RedSignal(x) for the same object, which is logically inconsistent for a real traffic controller.

- **Add a rule for emergency priority or ranking.**
- **Add lane/route conflict and collision-avoidance rules.**

• Represent traffic-light state consistently so Green and Red cannot be simultaneously active.

• Add rules for different emergency vehicles approaching the same intersection.

• Add time or location information to coordinate simultaneous vehicles.

CASE STUDY 3 – Agricultural Expert Reasoning System

Knowledge base:

- **P1: SoilDry → IrrigationNeeded**
- **P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod**
- **P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk**
- **Facts: SoilDry = True, CropWheat = True.**

(a) Convert P1, P2, P3 and facts into CNF

CNF requires every expression to be a conjunction of clauses, where each clause contains literals joined by OR.

- **P1: SoilDry → IrrigationNeeded ≡ ¬SoilDry ∨ IrrigationNeeded.**
- **P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod ≡ ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod.**
- **P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk ≡ ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk.**
- **Fact SoilDry = True gives clause SoilDry.**
- **Fact CropWheat = True gives clause CropWheat.**

For P2, the conjunction in the antecedent is moved into a disjunction by implication elimination: A ∧ B → C is equivalent to ¬A ∨ ¬B ∨ C. The same transformation is used for P3.

(b) Resolution refutation to prove ApplyDripMethod

Goal: ApplyDripMethod. Negate the goal and add ¬ApplyDripMethod to the CNF knowledge base.

- **C1: ¬SoilDry ∨ IrrigationNeeded.**
- **C2: ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod.**
- **C3: ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk.**
- **C4: SoilDry.**
- **C5: CropWheat.**
- **C6: ¬ApplyDripMethod (negated goal).**

Resolution steps:

- **Resolve C1 with C4 (SoilDry): derive IrrigationNeeded.**
- **Resolve C2 with IrrigationNeeded: derive ¬CropWheat ∨ ApplyDripMethod.**
- **Resolve that clause with C5 (CropWheat): derive ApplyDripMethod.**
- **Resolve ApplyDripMethod with C6 (¬ApplyDripMethod): derive the empty clause □.**

The empty clause proves that the negation of the goal is impossible. Therefore, ApplyDripMethod is logically entailed by the original KB.

(c) Effect of removing SoilDry

If SoilDry is removed, C4 is no longer available. Therefore, C1 cannot be used to derive IrrigationNeeded. Without IrrigationNeeded, P2 cannot derive ApplyDripMethod. Consequently, ApplyDripMethod is no longer logically guaranteed from the remaining facts.

CropWheat alone is insufficient to prove ApplyDripMethod. P3 also cannot be used to prove it because P3 has ApplyDripMethod in its conclusion and additionally requires ¬ApplyDripMethod as a premise. Thus, removing SoilDry breaks the reasoning chain and prevents the automatic recommendation of the drip method.

CASE STUDY 4 – Wumpus World Knowledge Agent

Given percepts and rules:

- **Stench at [1,2].**
- **Breeze at [1,1].**
- **Glitter at [2,2].**
- **Rule 1: Stench[1,2] → Wumpus is in a cell adjacent to [1,2].**
- **Rule 2: Breeze[1,1] → Pit is in a cell adjacent to [1,1].**
- **Rule 3: Glitter[2,2] → Gold is at [2,2].**
- **Rule 4: Wumpus[x,y] → ¬Pit[x,y] ∨ Safe[x,y].**

(a) Construct the agent's belief state

Assuming standard four-directional adjacency (up, down, left, right) and a 3×3 grid, the cells adjacent to [1,2] are [1,1], [1,3], and [2,2]. Therefore, the Stench indicates that the Wumpus may be in one of these cells.

The cells adjacent to [1,1] are [1,2] and [2,1]. Therefore, the Breeze indicates that a pit may be in [1,2] or [2,1]. The Glitter percept directly identifies the gold at [2,2].

- **Possible Wumpus locations: { [1,1], [1,3], [2,2] }.**
- **Possible Pit locations: { [1,2], [2,1] }.**
- **Gold location: [2,2].**

Rule 4 adds a constraint: if a particular cell contains the Wumpus, that cell cannot simultaneously contain a pit under the stated rule, and it is considered safe with respect to the pit condition. The percepts alone do not identify one unique Wumpus or pit cell; they narrow the candidates.

(b) Forward chaining to identify possible locations

- **Fact Stench[1,2] is present → infer Wumpus is adjacent to [1,2]. Candidate set = {[1,1], [1,3], [2,2]}.**
- **Fact Breeze[1,1] is present → infer Pit is adjacent to [1,1]. Candidate set = {[1,2], [2,1]}.**
- **Fact Glitter[2,2] is present → infer Gold[2,2].**
- **The agent records these possibilities in its knowledge base rather than claiming a single location without additional evidence.**

Therefore, the current percepts provide partial knowledge. Additional percepts from neighboring cells would be needed to eliminate candidates and determine the exact Wumpus and pit locations.

(c) Evaluation of the path [1,1] → [1,2] → [2,2]

The proposed path should be evaluated using the knowledge inferred from the percepts. Cell [1,1] has a Breeze, so there is a possible pit in one of its adjacent cells, including [1,2]. Cell [1,2] itself has a Stench, indicating a possible Wumpus in one of its adjacent cells, including [2,2].

Therefore, the path cannot be declared completely safe from the given information alone. The final cell [2,2] contains the gold, but it is also a candidate Wumpus location because it is adjacent to the Stench cell [1,2].

Logical conclusion: the path reaches the gold location, but the KB does not prove that every cell on the path is safe. A rational agent should seek additional percepts or use additional Wumpus World rules before committing to the path. This demonstrates how logical inference helps an intelligent agent distinguish known facts from possible hazards.

Final Answers at a Glance

Case Study	Main Conclusion
1 – Medical Diagnosis	Flu(PatientA) and Pneumonia(PatientA) are derived; Pneumonia is proved by backward chaining.

2 – Traffic Agent	**θ={x/Ambulance}; Proceed(CarA) is proved by resolution; extra rules are needed for simultaneous emergency coordination.**
3 – Agriculture	**ApplyDripMethod is proved by resolution; removing SoilDry breaks the inference chain.**
4 – Wumpus World	**Wumpus candidates: [1,1], [1,3], [2,2]; Pit candidates: [1,2], [2,1]; Gold is at [2,2].**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand